

Bhupesh Cholake

Software Engineer | Python, AI Applications and Automated Testing

Nashik, Maharashtra, India | +91 93091 72962 | ✉ work.bhupesh@gmail.com

🌐 linkedin.com/in/thebhupesh | 🐙 github.com/RealBhupesh | 🌐 bhupeessh.in

Professional Summary

Software engineer with Python development experience and merged open-source contributions involving automated tests, deterministic fixtures and CI checks. Built AI applications and contributed across unfamiliar codebases, with a focus on reproducible behavior, debugging and software reliability.

Education

Bachelor of Engineering in Artificial Intelligence and Data Science

CGPA: 8.6/10

Relevant coursework: Machine Learning, Data Structures and Algorithms, Business Intelligence

Experience

Physicore Engine Pvt Ltd

June 2026 - August 2026

Founding Engineer (Internship)

- Developed a mesh-based physics solver in Python, translating mathematical models into browser-based simulation workflows.
- Built responsive interfaces with JavaScript, HTML and Tailwind CSS, connecting formula controls to simulation workflows.
- Owned development across solver logic, product UX and rapid technical iteration.

Independent

Project-based

Freelance Software Developer

Canada-based client

- Launched a conversion-focused website and SEO foundation for an automotive-detailing business, contributing to an estimated \$2,000 in revenue within three weeks.
- Automated order management and customer email workflows from the ground up, reducing manual coordination for the business.

Open-Source Contributions

Open-Source Contributor

March 2026 - Present

Ruby, Rails, Minitest, Elasticsearch, JavaScript, React, Rust, Python, Pytest, Ruff, GitHub Actions

- Migrated 15 **Gumroad** PriceCheckerService RSpec examples to fixture-backed Rails Minitest; contributed deterministic Elasticsearch isolation and cache-behavior tests. The final upstream version contained 42 assertions and ran in 17.0s versus 137.1s in a same-host benchmark. Merged PR #6609
- Introduced Ruff linting and GitHub Actions CI enforcement for the **Kalshi GenAI Trading Bot**; refactored the Python application and test suite to pass automated quality checks. Merged PR #13
- Integrated authoritative Kalshi market and series settlement rules into Grok prompts using API wrappers and caching; added 200+ lines of tests for the integration. Merged PR #14
- Added plain-text note support and agenda/task improvements to **Lokus**; delivered parser, editor, calendar and task-management changes across 25 files with automated tests. Merged PR #439

Selected Projects

AI Fitness & Posture Coach

github.com/RealBhupesh/fitness-coach-final

Next.js, React, TypeScript, PostgreSQL, Drizzle ORM, MediaPipe, Groq, OpenAI

- Built a capstone AI fitness platform with personalized plans, conversational coaching and progress tracking.
- Engineered a real-time MediaPipe pipeline to evaluate form, count repetitions, score movement and deliver posture feedback.
- Implemented API authentication, rate limiting and AI-provider fallback; added automated tests using Vitest and Playwright.

Pintu: AI-Powered Discord Community Bot

github.com/RealBhupesh/pintu-bot - 50 stars

Python, Discord.py, OpenRouter, Groq, Spotify API, FFmpeg

- Built a Discord bot with AI assistance, automated moderation, music, community analytics and configurable workflows.
- Implemented AI summaries and a persistent virtual economy spanning markets, property, loans, taxes, auctions and trading.

Nestly: AI-Powered Real Estate Marketplace

Full-Stack Project

React, TypeScript, TanStack Start, Supabase, PostgreSQL, Gemini, MapLibre, Zod

- Built authentication, property publishing, advanced search, interactive maps, saved listings, comparisons and owner dashboards.
- Integrated Gemini search and image analysis; added messaging, tour scheduling, alerts, analytics and row-level security.

Technical Skills

Languages: Python, TypeScript, JavaScript, SQL; Ruby (open-source contributions)

Testing & Reliability: Pytest, Minitest, Vitest, Playwright, test fixtures, Ruff, GitHub Actions (CI), Git

AI & Computer Vision: OpenAI API, Groq, Gemini, OpenRouter, MediaPipe

Applications & Data: React, Next.js, Node.js, REST APIs, PostgreSQL, Supabase, Drizzle ORM

Tools: Codex, Claude Code, Cursor, Hermes Agent, OpenClaw, Notion, Framer

Training

Ethnus: Selected for S.Dot, an advanced training cohort for high-performing students.

Talent Battle: Aptitude, logical reasoning, data structures and algorithms training.